

align: Scriptable Solid-Propellant Grain Burnback, Validated Against Closed Forms and a Published Motor

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1. Introduction

A solid rocket motor's pressure and thrust follow the area of its burning surface, and that area changes as the grain burns back. Computing it for a three-dimensional grain such as a finocyl is the job of a burnback code. **align** is a fork of the open-source burnback-3d solver [1] that makes that computation scriptable end to end:

- **A command line for the solver.** burnback-3d is a Qt desktop application; mesh import, boundary conditions, the run and the export are done by hand. **align** runs the same solver kernels from a command line, with no Qt and no display, on Linux, Windows and macOS.
- **A pipeline around it.** A grain defined by a few numbers is turned into CAD (build123d), meshed (Gmsh), solved (**align**), reduced to burn area against web, corrected for mesh error, and passed to a lumped chamber model for pressure and thrust.
- **Checks that can be re-run.** Closed-form grains, a mesh sensitivity study, an independent fast-marching solution and a published motor are scripts in the repository.

The motivation is control by a language model. An LLM agent cannot reliably click through a GUI or judge a contour plot, but it can write a grain as numbers, run a command, read a CSV and compare it with a formula. Every stage here takes text and returns text, none of them prompts, and the checks that say whether an answer is right are themselves commands. The worked example and this report were produced that way.

2. Method

Burn-time field. With a burn rate that is uniform over the surface, the flame reaches a point $\mathbf{x}$ of the propellant after burning a web $u(\mathbf{x})$ that satisfies the eikonal equation

$$|\nabla u| = 1, \quad u = 0 \text{ on the initial burning surface.} \quad (1)$$

burnback-3d reaches this field on a tetrahedral mesh by marching $\partial u / \partial \tau = 1 - |\nabla u| + \varepsilon D(u)$ in pseudo-time τ to a steady state, where D is a diffusive term that keeps the scheme stable. **align** runs these kernels unchanged. Boundary types are the solver's own: *inlet* for the initial burning surface, *outlet* for the case wall and inhibited faces, *symmetry* for the cut planes of the smallest repeating wedge, which is all that is meshed.

Burn area. The burning surface at web w is the iso-surface $u = w$. Its area $A(w)$ is summed over the mesh by marching tetrahedra, exactly for the piecewise-linear field, and multiplied by the number of wedges.

Mesh error and its removal. The diffusive term makes the scheme first-order accurate: the computed surface lags the true one by an amount proportional to the element size h , more where the front is strongly curved. On a practical mesh this shifts $A(w)$ by a few percent (section 4). Because the error is so nearly linear in h , two meshes suffice to remove its leading term by Richardson extrapolation. Extrapolating A at fixed w would smear every kink of the curve, since the two meshes place a kink at different webs. The pipeline instead extrapolates the web at a fixed fraction f of propellant burned,

$$w_0(f) = \frac{r w_{h/r}(f) - w_h(f)}{r - 1}, \quad r = h_{\text{coarse}} / h_{\text{fine}}, \quad (2)$$

and differentiates the result, $A = V_p df / dw_0$, with V_p the meshed propellant volume. Kinks stay sharp and the propellant mass is conserved exactly.

Ballistics. A lumped chamber follows $dP/dt = (RT_c/V)[(\rho_p - \rho_g)A\dot{r} - PA_t/c^*]$ with $\dot{r} = aP^n$, free volume $V(w)$ from the burn table, and thrust $F = C_F PA_t$ for an ideal nozzle. A second, quasi-steady 1-D model resolves the flow along the port and adds erosive burning (section 5). Throat erosion, dynamic burning and the ignition transient are not modelled.

3. Validation against closed forms

Three grain families with closed-form burn area were run through the whole pipeline, each swept over its geometry (15 cases; case diameter 10 in):

- **Tube**, ends inhibited: $A = 2\pi(r_p + w)L$. Port-to-case ratio 0.2 to 0.8.
- **BATES**, both ends burning: $A = 2\pi(r_p + w)(L - 2w) + 2\pi[R^2 - (r_p + w)^2]$. Length-to-diameter ratio 0.5 to 1.5.
- **Star**, ends inhibited: the classical two-phase perimeter of a star with flat flanks and a filleted tip. Number of points 5 to 8, angular fraction 0.5 to 0.9, point angle 60° to 100° . The formulas were first checked against a numerical offset of the star outline (within 0.06%) and against the CAD volume (to six digits).

Table 1 gives the difference from the closed form over the first 95% of the web. The last 5% is excluded because the burn area is discontinuous at burnout, which any mesh spreads over about one element; the burnout web is compared instead. Figures 1 to 3 show the curves.

Table 1: Extrapolated burn area against the closed form, ignition to 95% of the web. “Fine mesh” is the same comparison for the finer of the two meshes without extrapolation.

Family	Case	Tetrahedra	Max (%)	RMS (%)	Fine mesh RMS (%)	Burnout web (%)
Tube	port / case = 0.2	404 497	0.16	0.14	0.21	+0.04
Tube	port / case = 0.4	353 854	0.15	0.14	0.09	+0.03
Tube	port / case = 0.6	270 593	0.14	0.13	0.05	+0.02
Tube	port / case = 0.8	153 053	0.14	0.14	0.02	+0.00
BATES	L / D = 0.5	439 570	0.79	0.14	0.36	+0.57
BATES	L / D = 1	875 969	0.18	0.09	0.30	-0.01
BATES	L / D = 1.5	1 311 133	0.23	0.12	0.27	-0.01
Star	N = 5	730 709	0.20	0.13	0.40	—
Star	N = 6	600 182	0.12	0.04	0.39	—
Star	N = 7	488 962	0.16	0.07	0.40	—
Star	N = 8	423 514	0.27	0.14	0.41	—
Star	angular fraction 0.5	547 546	0.25	0.19	0.30	—
Star	angular fraction 0.9	678 836	0.19	0.11	0.48	—
Star	point angle 60°	616 484	0.26	0.14	0.30	—
Star	point angle 100°	563 847	0.26	0.20	0.52	—

The extrapolated burn area is within 0.27 % of the closed form in every case but one, and within 0.20 % RMS in all of them. The exception is the shortest BATES grain, whose two end faces meet at burnout: its largest difference, 0.79 %, is in the last few percent of the compared range, next to that discontinuity. The agreement is not exact. A residual of about 0.1 % remains after extrapolation, and for the thin-web tube it is larger than the already small error of the fine mesh alone.

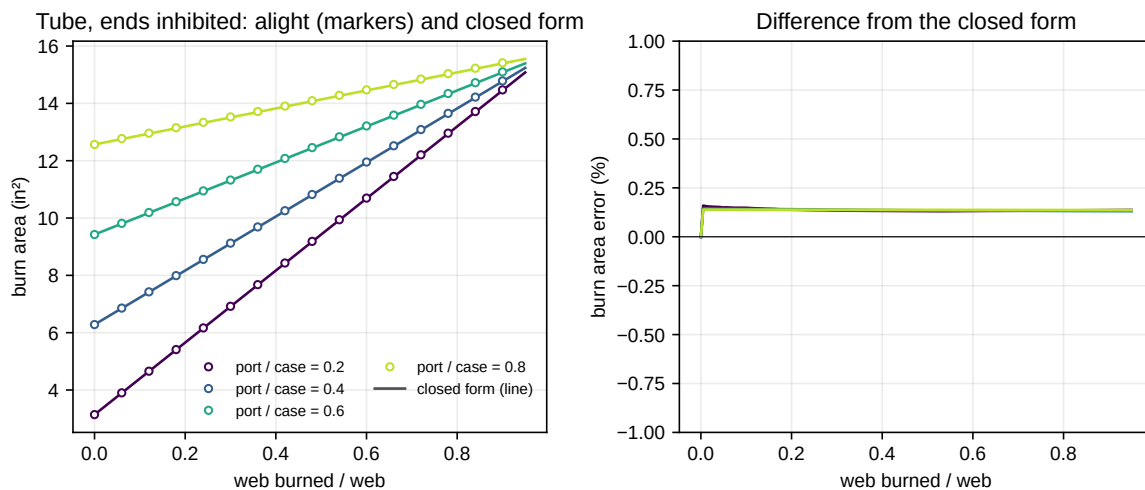


Figure 1: Tube grains: burn area and its difference from the closed form.

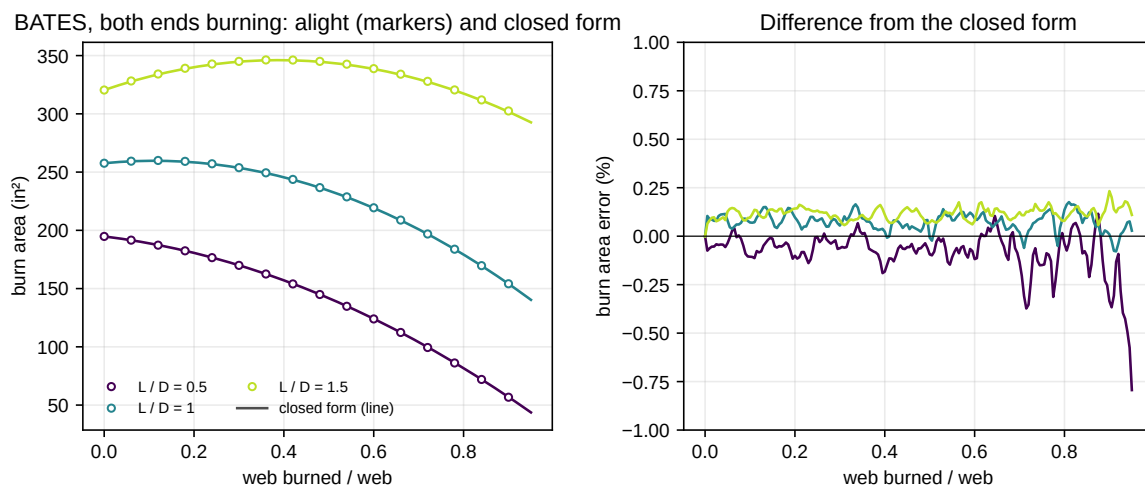


Figure 2: BATES grains.

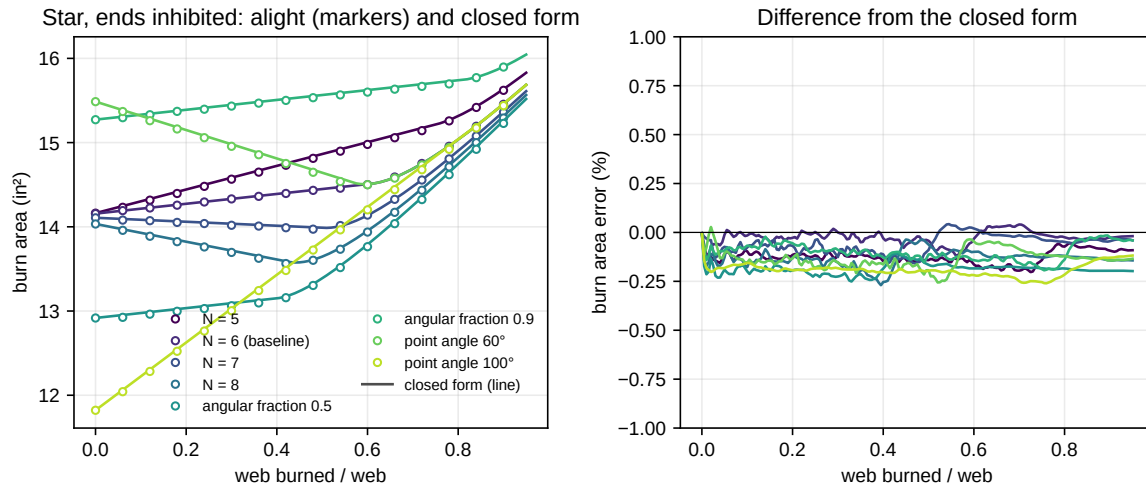


Figure 3: Star grains. The comparison ends where the star's arcs reach the case; the slivers that burn after that have no simple closed form.

4. Mesh sensitivity on a finocyl grain

The worked example is a finocyl grain: 10 in outer diameter, 4 in port, 60 in long, six 1 in fins with semicircular tips reaching a 4 in radius over the aft 20 in, case-bonded with both ends inhibited. It was solved on seven meshes, from 0.4 in to 0.05 in elements (24 thousand to 9.8 million tetrahedra in the 30° wedge), and each neighbouring pair was extrapolated. The propellant is an aluminized AP/HTPB composite with representative values ($\rho = 1800 \text{ kg/m}^3$, $\dot{r} = 0.35 \text{ in/s}$ at 1000 psi, $n = 0.35$, $c^* = 1580 \text{ m/s}$, $\gamma = 1.14$), and the throat (2.70 in) is sized for a 1000 psi peak.

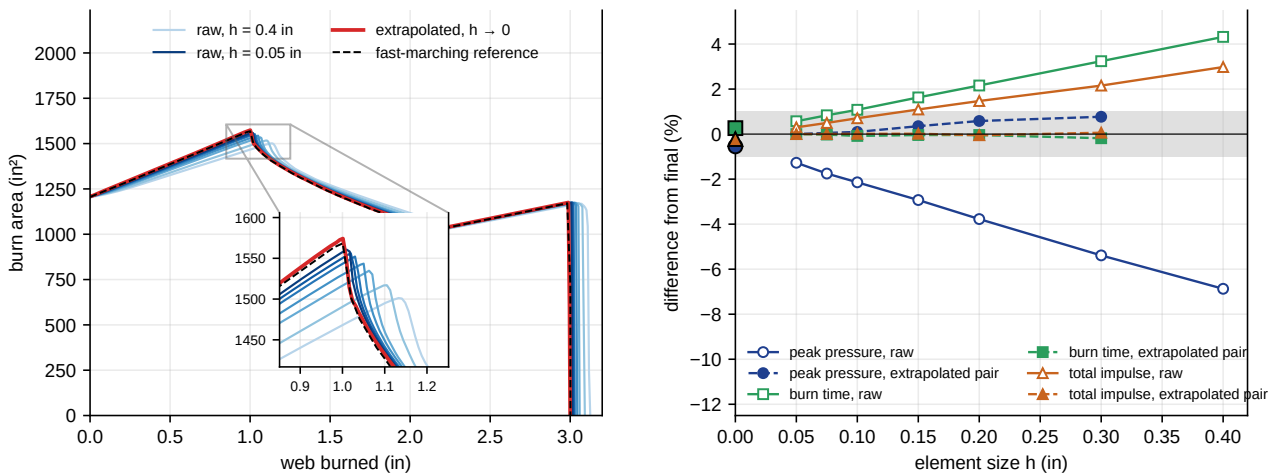


Figure 4: Left: burn area for the seven meshes, the extrapolated result and the fast-marching reference. Right: peak pressure, burn time and total impulse against element size, as difference from the final result; symbols at $h = 0$ are the reference, the band is $\pm 1 \%$.

- **Raw results depend on the mesh.** Peak pressure spans 5.6 % over the seven meshes and still moves 0.49 % at the last refinement. The error falls linearly with h : fitted orders are 0.81 for peak pressure, 0.98 for burn time and 1.04 for burnout web.
- **Extrapolated results do not.** For every pair from 0.2/0.15 in down, peak pressure, burn time and total impulse stay within 0.35 %; the last two pairs differ by at most 0.05 %.
- **An independent method agrees.** The distance from the initial port was computed on a 0.035 in Cartesian grid by fast marching (scikit-fmm), with burn area from marching cubes (scikit-image). The final result differs from it by 0.55 % in peak pressure, 0.26 % in burn time and 0.22 % in total impulse. The closed-form distance on the same grid agrees with the fast-marching one within 0.16 %.
- **Solver settings.** The CFL number and the convergence tolerance do not change the converged field. The diffusive weight changes the raw result by about 6 % but the extrapolated one by about 1 %; weights below about 0.75 diverge.

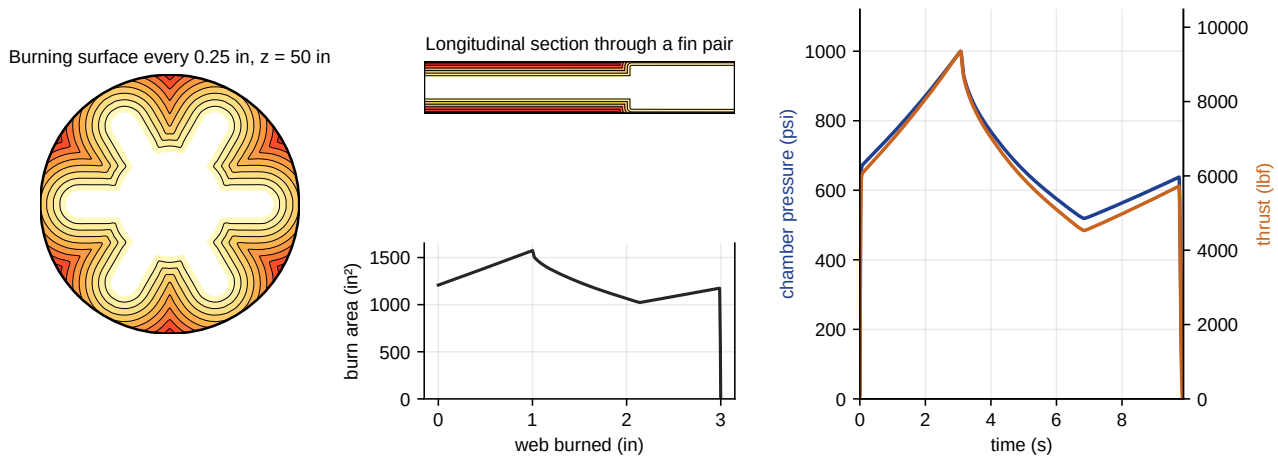


Figure 5: Finocyl example: burning surfaces, burn area, chamber pressure and thrust. Burn area peaks when the fin tips reach the case (web 1.00 in), falls as the slots burn out, and rises again in the plain port until burnout (web 3.00 in). Peak thrust 9355 lbf, burn time 9.82 s, total impulse 61 559 lbf s, propellant mass 110 kg.

5. Comparison with a published motor

NAWC tactical motor no. 6 is a reduced-smoke motor fired at China Lake [3] and simulated by Willcox et al. [2] with the Rocgrain three-dimensional burnback code. Its grain (5 in diameter, 69.4 in long) has a cylindrical bore at the head end that changes over tapered transitions into a six-slot star. Figure 7 of [2] dimensions it at seven stations, which were entered as a station-defined grain; the 0.213 in rounding where a slot meets the bore was left out, and the aft face was taken as burning because only the forward face is marked inhibited. The propellant is from Table 3 of [3]: 1.80 g/cm³, 0.678 cm/s at 6.9 MPa, exponent 0.36. Neither source states c^* ; it was derived from the tabulated chamber speed of sound (1083 m/s) with an assumed $\gamma = 1.2$, giving 1524 m/s. The throat is 1.84 in.

Figure 6 compares chamber pressure with Figure 10 of [2], read from the paper's vector graphics. The like-for-like reference is the paper's 0-D simulation with the strand burn rate, which makes the same modelling assumptions as this pipeline.

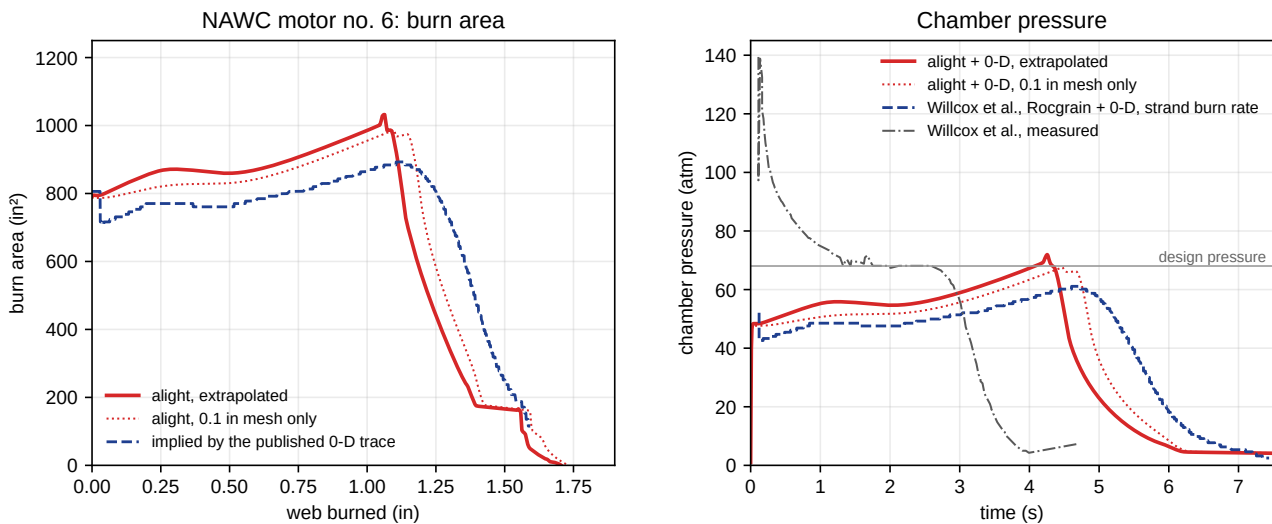


Figure 6: NAWC motor no. 6. Left: burn area from alight, and the burn area implied by the published 0-D trace (quasi-steady inversion with the strand burn rate). Right: chamber pressure.

What agrees. Three checks do not depend on the unknown c^* or on how sharply a code resolves the burn area. (i) The initial burn area: 792 in² from alight against 806 in² implied by the first point of the published trace (2%). (ii) The pressure integral, which a mass balance fixes at propellant mass times c^* over throat area: 297 against 307 atm s (3%); this also shows the derived c^* is close to the one behind the published run (1571 m/s for the same propellant mass). (iii) The web burned when the pressure peaks, which marks the slot tips reaching the case: 1.06 against 1.10 in (4%; the slot-tip webs in Figure 7 are 1.085 to 1.133 in).

What does not. Between ignition and the peak, alight's pressure is 14% above the published simulation, it peaks higher and earlier (72 atm at 4.25 s against 61 atm at 4.60 s) and it tails off sooner (half the peak at 4.7 s against 5.7 s). The left panel shows why: the published trace implies a burn area about a tenth lower before the slot tips

reach the case and much higher after, with a rounded peak in place of a corner. Since the two start from the same area and release the same total, the difference is in how the area is distributed over the burn, not in the inputs. A lower c^* cannot explain it: lowering c^* until the plateaus match (to about 1400 m/s) would mean the published run burned 12 % more propellant than the grain holds.

The difference has the signature of burnback resolution. alight's own result on the coarse 0.1 in mesh alone, before extrapolation, moves about half-way to the published curve (7 % instead of 14 % on the plateau, peak at 4.47 s). The resolution of the published burnback is not reported, so this is an inference, not a finding. The measured trace, finally, is higher and much shorter than every simulation (plateau 68 atm, equal to the 1000 psi design pressure; burnout near 3.3 s). Willcox et al. attribute that to erosive burning in the long, narrow star section, which no 0-D model captures (it is taken up below); alight's plateau (59 atm at 3 s) is closer to it than the published 0-D run (51 atm) but the measurement does not test the burnback.

With erosive burning. The measured trace is the real test of this motor, and it needs physics the 0-D model lacks: the star section's port is barely larger than the throat at ignition (2.9 against 2.7 in²), so the gas reaches Mach 0.6 in the port and scours the surface. A quasi-steady 1-D model was therefore added (alight.ballistics1d). The port is divided into 70 axial stations; each has its own web burned and takes its burning perimeter and port area from the 3-D solution at that web, extrapolated between the two meshes station by station. At every time step the steady flow along the port is solved with mass addition, a momentum balance for the static pressure and a choked nozzle, and the local burn rate is the pressure term plus the Lenoir–Robillard erosive term in the form [2] quotes from the Solid Performance Program,

$$\dot{r} = aP^n + \alpha \frac{G^{0.8}}{f(D_h)^{0.2}} e^{-\beta \dot{r}_{\rho p}/G}, \quad \beta = 53, \quad (3)$$

with G the local mass flux and $f(D_h)$ the program's fit in the hydraulic diameter. α follows from Lenoir and Robillard's heat-transfer expression; the flame temperature (2986 K) is from [3], and three inputs that neither source gives were set to typical values before the comparison was made: gas viscosity 8.5×10^{-5} Pa s, specific heat of the solid 1500 J/(kg K), surface temperature 1000 K. That gives $\alpha = 1.03 \times 10^{-5}$ in SI units. Nothing was fitted to the firing.

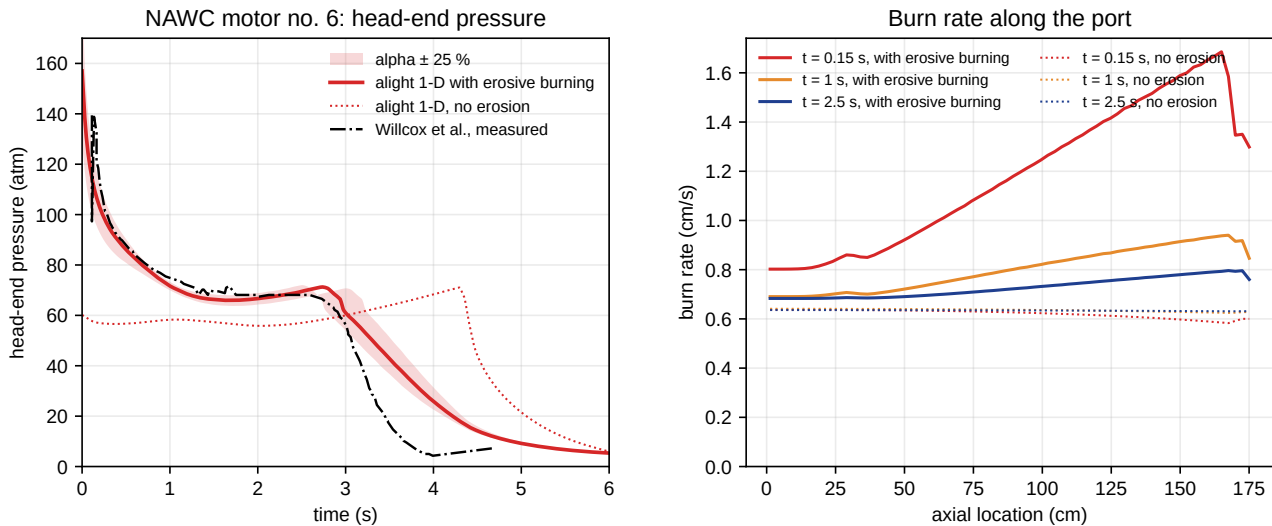


Figure 7: NAWC motor no. 6 with the 1-D model. Left: head-end pressure with and without erosive burning against the measured trace; the band is $\alpha \pm 25\%$. Right: burn rate along the port at three times.

Figure 7 shows the result. Over the quasi-steady part of the firing (0.3 to 2.8 s) the predicted head-end pressure differs from the measurement by -1.7 % on average and 3.7 % RMS: 86 against 88 atm at 0.5 s, 72 against 75 at 1 s, 67 against 68 at 2 s. Without erosion the same model is 21 % low and has the wrong shape: progressive where the firing is regressive. The model reproduces the regressive trace because erosion is strongest at ignition (up to 2.6 cm/s near the nozzle end against 0.6 cm/s without) and fades as the port opens. The initial pressure is 157 atm against a measured peak of 139 atm; the model has no ignition transient, so only the level is comparable. The weakest point is the tail-off: the predicted pressure falls through half its plateau at 3.77 s, the measured one at 3.27 s. Each station regresses by its own web as under uniform burning, which is least accurate once the slots have burned through unevenly; throat erosion, which is not modelled, would also shorten the tail. The result is moderately sensitive to the assumed properties: changing α by $\pm 25\%$ moves the RMS difference to 6.4 % and 2.8 %, and using the plain hydraulic diameter as the length scale gives 4.9 %.

6. Portability

alight is plain C++17 with one header-only dependency and builds with CMake. It was cross-compiled for Windows and run under Wine 10: on a 6,292-node mesh the Windows and Linux executables converged in the same number

of iterations and their burn-time fields agree to 1.4×10^{-14} mm. The pipeline's test suite (CAD, mesh, solver, burn area) also passes with the Windows executable substituted for the Linux one. A continuous-integration job builds `align` natively on Linux, Windows and macOS and runs the pipeline tests on Linux and Windows. The Python pipeline uses `numpy`, `scipy`, `matplotlib`, `build123d` and `gmsh`, all of which publish Windows wheels, and calls nothing platform-specific.

7. Summary and limits

- The burnback-3d solver, run from a command line and wrapped in a scripted pipeline, reproduces closed-form burn area within about 0.27 % for tube, BATES and star grains over 15 geometries.
- Its raw results are first-order in element size and shift by several percent between practical meshes. Extrapolating two meshes at fixed fraction burned removes that: results then agree within 0.35 % across mesh pairs and within 0.6 % of an independent fast-marching solution.
- For NAWC motor no. 6, align agrees with the published three-dimensional burnback simulation in initial burn area (2 %), pressure integral (3 %) and web at peak pressure (4 %), but its pressure is 14 % higher before the peak and it tails off sooner. The published curve implies a smoother burn-area history than the tabulated geometry gives at mesh convergence.
- With a 1-D flow model and Lenoir–Robillard erosive burning, the predicted head-end pressure of the same motor is within 3.7 % RMS of the measured trace over 0.3 to 2.8 s, against 23 % without erosion, with no parameter fitted to the firing; three transport properties were assumed and the tail-off is about half a second late.
- Limits: the burnback itself assumes a uniform burn rate over the surface; the 1-D model applies erosive burning station by station on top of that solution. No throat erosion, dynamic burning or ignition transient; propellant values in the worked example are representative, not a qualified formulation. No Solid Performance Program (SPP) case could be reproduced: SPP is export-controlled, and its public manual [4] gives pressure traces for its validation motors but not their grain inputs.

References

- [1] iff, *burnback-3d*, <https://codeberg.org/iff/burnback-3d>.
- [2] M. A. Willcox, M. Q. Brewster, K. C. Tang, D. S. Stewart and I. Kuznetsov, "Solid Rocket Motor Internal Ballistics Simulation Using Three-Dimensional Grain Burnback," *Journal of Propulsion and Power*, Vol. 23, No. 3, 2007, pp. 575–584. doi:[10.2514/1.22971](https://doi.org/10.2514/1.22971).
- [3] F. S. Blomshield, *Pulsed Motor Firings*, NAWCWD TP 8444, Naval Air Warfare Center Weapons Division, China Lake, CA, 2000. DTIC ADA382239.
- [4] D. E. Coats et al., *A Computer Program for the Prediction of Solid Propellant Rocket Motor Performance*, AFRPL-TR-75-36, Vols. I–II, 1975. DTIC ADA015140, ADA015141.